

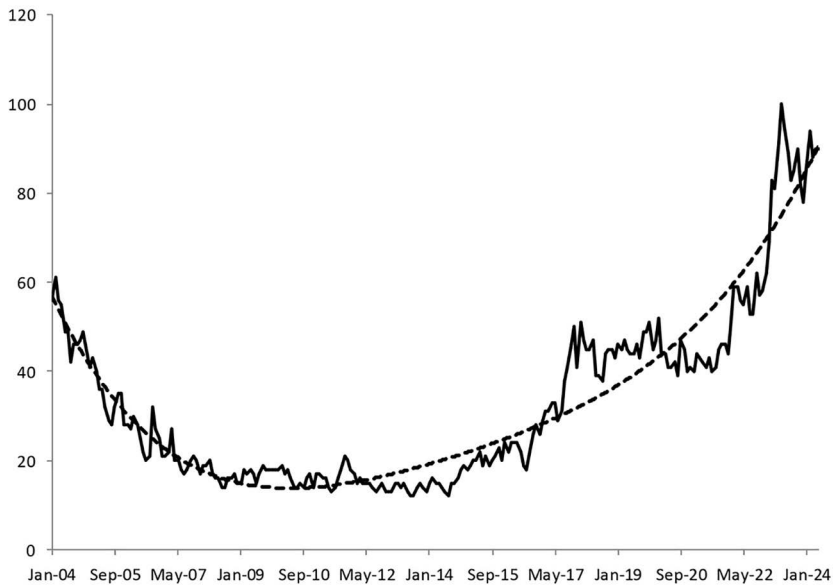
1. Artificial Intelligence: Definition, Origin and Evolution

1.1 DEFINITION OF ARTIFICIAL INTELLIGENCE

Artificial intelligence (AI), also but rarely known as computational intelligence, is intelligence displayed by computers and other machines, as opposed to the natural intelligence displayed by humans and animals. AI has attracted the interest of intelligent humans for hundreds of years, long before the 1950s when Alan Turing, a British mathematician and computer scientist, wondered if machines could think like humans. Interest in AI pre-dates the coining in the 1950s of the term “artificial intelligence” by John McCarthy, an American computer and cognitive scientist. Since the second half of the twentieth century, human intelligence has been able to create artificial intelligence in the form of machines that can learn and apply learning in ways that only humans could do, or so it was believed.

Today, we interact with AI through social media, online shopping, and search engines. We see AI in action every day, even though most users do not realise that they are using and interacting with AI. AI can be seen in, among others, digital assistants (Siri, Alexa, etc.), self-driving cars, navigation apps, social media algorithms, targeted advertisements, search engines, text editing, and gaming. Businesses use AI for a variety of purposes, including the automation of routine and repetitive tasks, fraud detection, and for obtaining valuable insights from the analysis of massive data sets, not to mention customer service, marketing, finance, and operations. In short, AI has become an integral part of our lives, whether we realise that or not, and this is why people have been increasingly searching for the meaning of AI by looking for a definition or description of AI. Figure 1.1 displays an index of Google searches for “artificial intelligence” relative to the peak of 100, which was reached in May 2023.

Even though AI has become rather ubiquitous, it is not easy to define because of its interdisciplinary and multifaceted nature and the diversity of its applications. In essence, however, AI refers to the ability of machines to perform tasks that typically require human intelligence, ranging from simple data analysis to decision-making in complex situations. An alternative term for artificial intelligence is “non-human intelligence,” a notion that can be traced back to



Data source: Google Trends.

Figure 1.1 An index of the frequency of Google searches for “artificial intelligence”

Greek philosophers. Sometimes, the term “human-like intelligence” is used to describe how machines can acquire and use human-like abilities such as deductive reasoning, inductive reasoning, mathematical reasoning, oral/written comprehension, and oral/written expression (for example, Korteling et al., 2021; Fleishman and Quaintance, 1984).

Before defining AI, perhaps it is useful to understand the meaning of the words “artificial” and “intelligence.” According to Dictionary.com, the word “artificial” means “made by human skill” or “produced by humans”. The word “intelligence” means “the capacity, especially of a particular person or animal, for learning, reasoning, understanding and similar forms of mental activity.” By combining the two words, artificial intelligence has two definitions in Dictionary.com. The first is that AI is “the capacity of a computer, robot, programmed device, or software application to perform operations and tasks analogous to learning and decision making in humans, such as speech recognition and question answering.” The second definition is “the branch of computer science involved with the design of computers, robots, programmed

devices and software applications having the capacity to imitate human intelligence and thought.”

Let us go back to the definition of intelligence on its own. Intelligence is defined by Manning (2020) as “the ability to learn and perform suitable techniques to solve problems and achieve goals, appropriate to the context in an uncertain, ever-varying world.” It follows that intelligence encompasses the abilities to (i) solve problems, (ii) use knowledge to devise novel solutions to problems, (iii) learn something and improve upon it, and (iv) make a decision. Michael Bennett (2024) lists the terms used to describe human intelligence as smartness, understanding, brainpower, ability to reason, sharpness, and wisdom. However, he argues that “hundreds of definitions have been created, but for much of that time, intelligence has meant a biopsychological capacity to acquire and apply knowledge and skills.”

The pioneers of AI have come up with various definitions of intelligence. McCarthy (2007) proposed the definition that “intelligence is the computational part of the ability to achieve goals in the world,” suggesting that “varying kinds and degrees of intelligence occur in people, many animals and some machines.” For Newell (1990), an intelligent system “operates in real-time; exploits vast amounts of knowledge; tolerates erroneous, unexpected and possibly unknown inputs; uses symbols and abstractions; communicates using some form of natural language; learns from the environment; and exhibits adaptive goal-oriented behaviour.” Kurzweil (1990) argues that “there appears to be no simple definition of intelligence that is satisfactory to most observers, and most would-be definers of intelligence end up with long checklists of its attributes.”

Perhaps one way to understand artificial intelligence is to compare it with human intelligence. While human intelligence requires the use of the brain, memory, and cognitive abilities, AI relies on the data provided by humans. The two kinds of intelligence differ in many respects. In terms of processing, AI is based on algorithms and mathematical models provided by humans, whereas human intelligence is based on cognitive processes and biological structures. In terms of learning, AI is based on data and feedback loops, whereas human intelligence is based on experience, intuition, and creativity. They also differ in terms of the speed of executing tasks because AI can process data and perform tasks much more quickly than the most intelligent humans can, but this does not make machines better than humans because humans can make complex decisions quickly. In terms of emotions, machines are devoid of emotions and empathy, whereas humans are capable of feeling emotions and empathy. Likewise, machines have no moral code or conscience, whereas humans (albeit not all of them) are guided by a moral code and conscience when they make decisions. In terms of creativity, machines have a limited ability to be creative or think “outside of the box,” whereas humans can be creative, imaginative,

and innovative. Last but not least, machines (unlike humans) are not exposed to physical limitations and can operate continuously, subject to maintenance. Humans, on the other hand, have limited physical capabilities and require rest. Unlike humans, machines do not mix business with pleasure.

John McCarthy, the computer scientist who coined the term “artificial intelligence,” defined AI very simply as “the science and engineering of making intelligent machines” (Manning, 2020). A slightly more modern definition of AI is that it is a “broad branch of computer science concerned with creating machines that can learn, make decisions, and perform tasks to a human-like level” (Tableu, 2024). Otherwise, it is defined as a “specialty within computer science that is concerned with creating systems that can replicate human intelligence and problem-solving abilities.” IBM (2024) defines AI as “technology that enables computers and machines to simulate human intelligence and problem-solving capabilities.” Russell and Norvig (2021) define AI as a “field of research in computer science that develops and studies methods and software which enable machines to perceive their environment and uses learning and intelligence to take actions that maximize their chances of achieving defined goals.”

AI may be defined in terms of the ability of AI systems to think or act like humans or rationally. Bellman (1978) defines AI as “the automation of activities associated with human thinking, which include decision making, problem solving, and learning.” This is a description of systems that think like humans. A definition that describes systems that think rationally is that AI is “the study of the computations that make it possible to perceive, reason, and act” (Winston, 1992). Kurzweil (1990) defines AI as “the art of creating machines that perform functions that require intelligence when performed by people,” which is a description of systems that act like humans. Luger and Stubblefield (1993) define AI, in terms of systems that act rationally, as the “branch of computer science that is concerned with the automation of intelligent behaviour.”

A question that remains is the following: what is in a definition that motivates the desire to find a definition for AI? For example, is it informative to define a car as “an entity that is used to move people and goods from one place to another, propelled by an internal combustion engine that derives power from liquid fuel according to the laws of thermodynamics”? Is it informative to define water as a “chemical compound with a molecule that consists of two atoms of hydrogen and one atom of oxygen”? In both cases, the answer should be in the negative because we recognise cars and water when we see them. Even though AI is more opaque than cars and water, a definition of AI is not informative as long as we know that AI is about teaching machines how to think and operate like humans.

Rather than trying to understand what AI is by looking at various definitions, one can do that by observing a robot tasked with welding car parts on a

conveyor belt in a car factory, or a robot that beats chess grandmasters at their own game. If a layperson asks what artificial intelligence is, a good answer would be that it is the human endeavour that has created the annoying machine that talks to you when you call your bank, expecting to talk to a human being. It is also the human endeavour that has created the Google search engine, which takes a few seconds to give you thousands of results when you type “artificial intelligence.” This is the same product of AI that has significantly reduced the time it takes to write a book like this one, without having to visit a physical library. It follows that the statement “there has yet to be a universally accepted technical or legal definition of AI” is an unnecessary dramatisation of a trivial matter.

1.2 EVOLUTION OF AI: THE ORIGIN

The modern conception of AI emerged in the 1950s as a result of significant advancements in computing technology, as well as logic and mathematics. However, the origin of AI can be traced back to ancient inventors and thinkers who contemplated the possibility of creating artificial beings that have human-like abilities. Furthermore, without the groundwork that began in the early 1900s, it would not have been possible for early AI experts to come up with their innovative work in the 1950s. In this section, we examine the pre-1900 period.

The notion of AI goes back thousands of years to the times when ancient philosophers considered questions of life and death. Human history provides abundant stories about ancient inventors who produced mechanical devices that moved without human intervention. According to Encyclopaedia Britannica (2024), an automaton is “any of various mechanical objects that are relatively self-operating after they have been set in motion.” The term is also applied to a “class of electromechanical devices that transform information from one form into another on the basis of predetermined instructions or procedures.” Andrews (2023) describes seven early robots and automatons, which include the sixteenth-century mechanical monk, Al-Jazari’s floating orchestra, Archytas’ Dove, the silver swan, Jaquet-Droz’s three automatons, Vaucanson’s duck, and Da Vinci’s knight. Encyclopaedia Britannica (2024) refers to the wooden pigeon of Archytas, who was a friend of Plato, as “among the earliest references” to an automaton and notes that more complete information about similar devices can be found in the writings of Heron of Alexandria (first century CE), who described devices propelled by water, falling weights, and steam.

Even in the Middle Ages of Western Europe (the period extending between the collapse of the Roman Empire in the fifth century CE and the period of the Renaissance), automatons were created by the likes of Roger Bacon (a medieval

English philosopher who was interested in experimental science) and Albertus Magnus (a German thinker who established the study of nature as a legitimate science within the Christian tradition). Both Bacon and Magnus are credited with constructing androids (humanoid robots or other artificial beings, typically made from a flesh-like material). More specifically, Bacon invented a talking head, whereas Magnus constructed an iron man. The French Gothic architect Villard de Honnecourt compiled a sketchbook in which he illustrated decorative mechanical objects for churches.

Renewed interest in the production of automatons emerged in the early sixteenth century, largely stemming from the influence of Eastern inventions, which arrived in Europe through trade with the Orient. Also influential was the translation from ancient Greek of the writings of Heron of Alexandria on mechanical objects. In the mid-fifteenth century, the use of coiled tempered-steel springs as a portable source of motion became widespread. It was used, for instance, to make table ornaments in the form of sailing ships. Among the more elaborate mechanical devices that became popular in the eighteenth century were mechanical pictures, in which windmills and other features sprang to life by means of hidden clockwork. In 1813, the Rochat brothers, who manufactured miniature singing birds, decided to open their own shop in Geneva, where they assembled their birds and shipped them to London (the centre of finance and commerce at that time) to be exported to the rest of the world. With the exception of a few works of Peter Carl Fabergé (a Russian goldsmith and jeweller who is best known for his Fabergé eggs), the production of costly artistic automatons virtually came to an end in the late nineteenth century and early twentieth century because of the near extinction of skilled craftsmen and the rich patrons who supported them.

1.3 EVOLUTION OF AI: GROUNDWORK (1900–1950)

The word “robot” emerged in the 1920s, not as a result of the work of a mathematician, engineer, or inventor, but rather out of the work of a Czech playwright who, in 1921, released the science fiction play “Rossum’s Universal Robots.” The play introduced the notion of “artificial people,” which he called “robots”—that was the first known use of the word. The story, according to the English translation of the play, goes as follows (Capek, 2019):

The play is set on an island somewhere on our planet, and on this island is the central office of the factory of Rossum’s Universal Robots. “Robot” is a Czech word meaning “worker.” When the play opens, a few decades beyond the present day, the factory has already turned out, following a secret formula, hundreds of thousands, and even millions, of manufactured workmen, living automatons, without souls, desires, or feelings. They are high-powered laborers, good for nothing but work. There are two grades, the unskilled and the skilled, and specially trained workmen

are furnished on request.... The rest of the play is magnificent melodrama, superbly portrayed, with the handful of human beings at bay while the unseen myriads of their own robots close in on them. The final scene is like Dunsany on a mammoth scale... Then comes the epilogue, in which Alquist, the company's builder, is not only the only human being on the island, but also the only one left on earth. The robots have destroyed the rest of mankind.

In 1929, a Japanese biologist, who was alarmed by Capek's play, built the first Japanese robot, *Gakutensoku*, which means "learning from the rules of nature." In an article published in the *Osaka Mainichi* newspaper, Makoto Nishimura (1883–1956) said that he was concerned because Capek's play convincingly portrayed "the emergence of a perverse world in which humans become subordinate to artificial humans" (Frumer, 2020). Nishimura believed that the character of artificial humans was determined by the intentions of their creators. He also believed that if humanoids were intended to be slave workers, taking orders from their human masters, this situation would inevitably lead to the emergence of an "exploited underclass that was bound to rebel," which is exactly what happened in the play. As a result, Nishimura was apprehensive about a dystopian scenario that he described as humanity "destroyed by the pinnacle of its creation." In response, he envisaged a utopian scenario by creating a conscientious, altruistic, and empathetic artificial human, a robot that was not a slave but a friend and a partner, even a role model. In 1926, Nishimura resigned his position as a professor at Hokkaido Imperial University, moved to Osaka, and devoted his time to the building of his ideal artificial human.

In 1949, Edmund Berkley, a computer scientist, published a book, *Giant Brains, or Machines That Think*, in which he compared the newer models of computers to human brains. In his description of the book, Berkley (1949) opined the following:

The subject of this book is a type of machine that comes closer to being a brain that thinks than any machine ever did before 1940. These new machines are called sometimes mechanical brains and sometimes sequence-controlled calculators and sometimes by other names. Essentially, though, they are machines that can handle information with great skill and great speed. And that power is very similar to the power of a brain.

Berkley envisaged the characteristics of the new machines that make them important: (i) they do the work of hundreds of humans for the wages of a dozen; (ii) they are powerful tools for the endeavour to obtain new knowledge; (iii) they can be used in science, business, government, and other activities; and (iv) the harder the underlying problem, the more useful these machines become. Two of these points pertain to the economics of AI, which is the

subject matter of this book. The first is that a profit-maximising firm can cut costs by employing machines rather than humans, which is one of the main issues discussed in this book and arguably the most adverse consequence of AI. The second is that these machines can be used across the economy in a wide range of applications, as we are going to see in Chapter 3.

1.4 EVOLUTION: BIRTH, EARLY DEVELOPMENT AND SETBACK (1950–1979)

The birth of AI is believed to have taken place during the period 1950–56. This period witnessed the publication of Alan Turing's influential book *Computer Machinery and Intelligence* (Turing, 1950), as well as the emergence of the term “artificial intelligence.” The same period also witnessed the development of symbolic AI systems, which relied on rule-based reasoning and symbolic manipulation of data. Researchers (such as John McCarthy, Marvin Minsky, and Herbert Simon) pioneered this approach to the building of machines that can indulge in logical reasoning and perform problem solving. Symbolic AI relies on explicit, human-readable symbols to represent knowledge and information, enabling logical manipulation and reasoning.

In the 1950s, Alan Turing laid the theoretical groundwork for computational thinking and machine intelligence, which made him influential in the development of theoretical computer science and a founder of AI. Turing (1950) raised the question of whether or not machines can think. He started by defining the terms “machine” and “think,” which he explained as follows:

I PROPOSE to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, ‘Can machines think?’ is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

Accordingly, he suggested that the new form of the problem could be described in terms of a game that he called the “imitation game,” eventually formulating the “Turing Test,” whereby a human interrogator tries to distinguish between the text responses generated by computers and humans. The Turing Test is used to determine whether or not a machine can demonstrate human intelligence, which happens if it can engage in a conversation with a human without being detected as a machine.

In 1952, a computer scientist named Arthur Samuel developed a program to play checkers, which was the first ever to learn the game independently. According to Knopp (2023), Samuel's "groundbreaking checkers program, developed in the early days of AI (on an IBM 701 computer), played a pivotal role in advancing the field of AI." The program involved one of the very first examples of non-numerical computation (early computers were intended and designed primarily for number crunching). Samuel went on to popularise the term "machine learning" as his checkers program highlighted the potential of algorithms to learn and improve through experience. He investigated two machine-learning procedures by using checkers, reaching the conclusion that "enough work has been done to verify the fact that a computer can be programmed so that it will learn to play a better game of checkers than can be played by the person who wrote the program," adding that "the principles of machine learning verified by these experiments are, of course, applicable to many other situations" (Samuel, 1959).

In the summer of 1956, John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon coined the term "artificial intelligence" in a proposal that they wrote to call for the convening of the famous Dartmouth Conference, which was held at Dartmouth College (McCarthy et al., 1955). The proposal started with the following introduction:

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College....The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans.

The Dartmouth Conference, held in Hanover (New Hampshire), brought together the intellectual power of those who were passionate about the emergent idea of designing intelligent machines. One of the pivotal outcomes of the conference was the conceptual exploration of AI, touching upon neural networks, theories of the human brain, machine self-improvement, abstraction, and creativity. The conference paved the way for a number of AI breakthroughs in subsequent years.

In 1956, Allen Newell, Cliff Shaw, and Herbert Simon were working at the Rand Corporation in Santa Monica, California, where they created the Logic Theorist (LT), which was the first program designed to perform automated reasoning. It has been described as "the first artificial intelligence program" because of its ability to mimic the problem-solving skills of a human being (for example, McCorduck, 2004). The Rand Corporation team used the program for proving theorems found in *Principia Mathematica*, a three-volume book on the foundations of mathematics that was originally published in 1910–1913

(Whitehead and Russell, 1963). Simon (1998) described Logic Theorist and the reason for using *Principia Mathematica* as follows:

LT was based on the system of Principia Mathematica, largely because a copy of that work happened to sit in my bookshelf. There was no intention of making a contribution to symbolic logic, and the system of Principia was sufficiently outmoded by that time as to be inappropriate for that purpose. For us, the important consideration was not the precise task, but its suitability for demonstrating that a computer could discover problem solutions in a complex nonnumerical domain by heuristic search that used humanoid heuristics.

The period from the late 1950s through the 1960s witnessed the invention of programming languages that are still in use, as well as the publication of books and screening of movies that explored the notion of robots, making AI a mainstream idea. In 1958, John McCarthy created List Processing (LISP) as the first programming language used for AI research, which is in popular use to this day. LISP became a common language for AI programming, partly because of the confluence of LISP and AI work at MIT and partly because AI programs that are capable of “learning” could be written in LISP (see, for example, Hemmendinger, 2024).

The 1960s started with the introduction in 1961 of the first industrial robot that was put to work on an assembly line in General Motors factories, designed specifically to do jobs deemed too dangerous for humans. In 1965, Edward Feigenbaum and Joshua Lederberg created the first “expert system,” which was a form of AI programmed to replicate the thinking and decision-making abilities of human experts. The system, called DENDRAL, was designed to carry out chemical analysis to help chemists identify unknown organic molecules. In 1966, Joseph Weizenbaum invented the first chatbot that used natural language processing (NLP) to converse with humans. In that same year, a Soviet mathematician, Alexey Ivakhnenko, proposed a new approach to AI that would later become known as “deep learning.”

The development of AI witnessed a setback in the 1970s. In 1973, a British applied mathematician named James Lighthill submitted a report to the British Science Council, known as the *Lighthill Report*, in which he presented a very pessimistic outlook for research on AI, stating that “in no part of the field have the discoveries made so far produced the major impact that was then promised” (Lighthill, 1973). Lighthill claimed that AI researchers had failed to address the issue of combinatorial explosion when solving problems within real-world domains. He also suggested that AI techniques may work within the scope of small problem domains, but the techniques are inadequate for solving more realistic problems (see, for example, Howe, 2007; Russell and Norvig, 2021). In 1979, however, the Stanford Cart became an early example

of an autonomous vehicle when it navigated successfully, and without human interference, a room full of chairs (see, for example, Earnest, 2012).

1.5 EVOLUTION: BOOM FOLLOWED BY SLUMP (1980–1992)

The 1980s saw a shift towards machine learning techniques, intended to enable computers to learn from data and make predictions without explicit programming. This period witnessed the emergence of algorithms such as neural networks, decision trees, and genetic algorithms, laying the groundwork for modern AI applications. The 1980s also witnessed rapid growth and rising interest in AI as a result of some notable breakthroughs and the expansion of government funding to support AI research. Deep learning and expert systems allowed computers to learn from their mistakes and make independent decisions.

Expert systems are computer systems that can emulate the decision-making abilities of humans. In 1980, the first commercial expert system, XCON, was launched, allowing customers to order computer systems automatically and validate the technical configurability of their orders. The system also provided guidance on the assembly of the hardware. Further developments include the advent of the autonomous drawing program, AARON, which is described by Garcia (2016) as “one of the longest-running, continually maintained AI systems in history.” In 1986, Ernst Dickmann demonstrated the first driverless car. And in 1987, Alacrity became the first managerial advisory system that was available commercially (see, for example, Cook and Sterling, 1989).

In 1984, the Association for the Advancement of Artificial Intelligence (AAAI) warned of a looming “AI Winter,” predicting that the funding of and interest in AI would diminish, which would hamper research. The reason for the loss of interest by the private sector, and also the public sector, was the perceived high cost-to-return ratio, which was a sign of the end of a hype cycle (see Chapter 8). Schuchmann (2019) thinks that the 1980s AI Winter came in 1988, but he calls it the second AI Winter, following the first AI Winter of 1973. In 1987, the market for LISP-based hardware collapsed owing to the emergence of cheaper and more accessible competitors that could run LISP software, including those offered by IBM and Apple. The shift in market conditions led to the failure of many specialised LISP companies. Computers made by Apple and IBM became more powerful than LISP machines and other expert systems, which were expensive to maintain. Still, the year 1988 witnessed the invention of a chatbot called Jabberwacky, which was capable of conducting conversations with humans. Jabberwacky’s new version, Cleverbot, performed so well in convincing humans that they were not talking to a machine that it passed the Turing Test spectacularly (see, for example, Dey, 2019).

1.6 EVOLUTION: RESURGENCE FOLLOWED BY EXPLOSION (1993–PRESENT)

The early 1990s witnessed significant leaps in AI research, including the introduction of the first AI system that could beat a reigning world champion in his own game (chess). In 1997, a computer system developed by IBM, Deep Blue, managed to beat the world chess champion, Garry Kasparov, in a highly-publicised match. Around the same time, Windows released speech recognition software (developed by Dragon Systems). Several inventions followed: Kismet, the robot that could simulate human emotions (2000), Roomba the vacuum cleaner (2002), Watson the Jeopardy winner (2011), and Siri the virtual assistant (2011).

An explosion in AI products has been witnessed since 2012, particularly the growth of applications utilising deep learning and big data (data sets that are too large or complex to be dealt with by traditional data-processing software). Unfortunately, one of these applications is AI-driven autonomous weapons (something like robots with machine guns deciding independently whom to shoot). This application became a new item on the wish list of generals, the military-industrial complex, and war-mongering politicians. In response, over 1,000 high-profile AI experts and leading researchers (including Elon Musk, Stephen Hawking, and Steve Wozniak) signed an open letter in 2015 warning of a “military artificial intelligence arms race” and calling for a ban on “offensive autonomous weapons.” The letter, presented at the International Joint Conference on Artificial Intelligence in Buenos Aires, described autonomous weapons as the “third revolution in warfare, after gunpowder and nuclear arms” (Gibbs, 2015).

However, the peaceful, welfare-enhancing AI devices kept on coming. Even though generative AI (which can be used to create new content, including audio, code, images, text, and videos) was introduced in the 1960s in chatbots, it was not until 2014, with the introduction of generative adversarial networks (GANs), that generative AI became capable of creating authentic images, videos, and audio of real people. In 2016, Sophia was created as the first “robot citizen” with a realistic human appearance and the ability to replicate and exhibit emotions. In 2017, Facebook programmed two AI chatbots to converse and learn how to negotiate. In 2020, OpenAI embarked on testing Generative Pre-trained Transformers (GPTs), starting with GPT-3, a model that was the first to create content almost indistinguishable from what can be produced by humans in their fields of expertise. In 2021, OpenAI developed DALL-E, which can process and understand images to the extent that it can produce accurate captions. Following its release in November 2022, ChatGPT reached over a million users in five days and 100 million users within two months,

making it the fastest-growing consumer app ever at the time (see, for example, Bianchi, 2024).

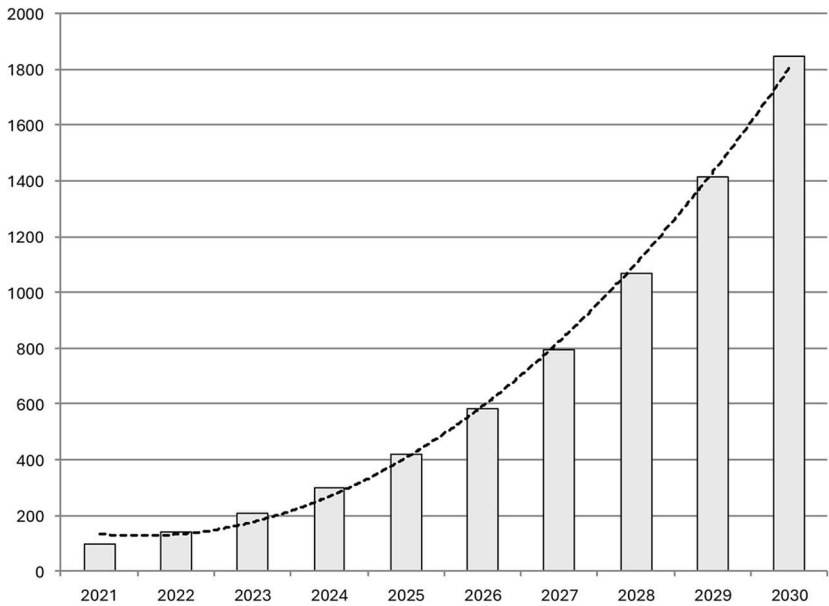
Following GPT-3, GPT-4 was released in March 2023, with enhanced abilities to write essays and articles and create art and music. GPT-5, which was released in September 2024, is significantly more advanced than its predecessor because of its ability to process and generate not just text, but also speech, images, code, and videos. Because it is so advanced, GPT-5 represents a significant step towards artificial general intelligence (AGI), a form of AI that surpasses human intelligence. Sravanthi (2024) notes that “while ChatGPT-5 is not an AGI system itself, it exhibits characteristics that align with the goals of AGI research.”

On 15 February 2024, OpenAI announced the release of Sora, an AI model that can create realistic and imaginative scenes from text instructions. On 22 February 2024, StabilityAI announced the introduction of Stable Diffusion 3, a text-to-image open model based on a similar architecture to Sora. Other GPT models have been developed by companies other than OpenAI. For example, EleutherAI has created a series of GPT foundation models, while Cerebras has developed seven models. Furthermore, companies across various industries have developed GPT models tailored to their specific needs. Examples include Salesforce’s EinsteinGPT for customer relationship management (CRM) and Bloomberg’s BloombergGPT for finance. By the time this book is published (sometime in 2025), new marvellous AI inventions will have appeared.

In 2023, the global AI market size was close to \$208 billion. By 2030, the market size is expected to reach nearly \$2 trillion, according to figures provided by Statista (Figure 1.2). The size of the AI market is measured by spending on hardware and services that enable firms and government departments to develop and deploy AI applications. The AI market consists of the markets for computer vision (applications that enable computers to interpret and understand digital images and video data), machine learning (algorithms that enable computer systems to learn from data), natural language processing (applications that enable computers to understand, interpret, and generate human language), robotics and autonomous technology (machines and systems that operate independently), and generative AI (used to generate content that is indistinguishable from content created by humans).

1.7 THE TRANSFORMATIVE AND DISRUPTIVE POWER OF THE AI REVOLUTION

A technology is disruptive if it starts by serving a niche market (a very specific segment of consumers who are likely to buy a particular product) and then expands to the mainstream (the average or typical consumer). Mann (2023) suggests that transformation and disruption are related by arguing that “a



Data source: Statista.

Figure 1.2 World market for AI (\$ billion)

disruptive technology is technology which can transform the way people and businesses operate.” AI is considered to be the fourth technological revolution, following the agricultural, industrial, and digital revolutions, all of which have produced transformative and disruptive effects on society and the economy. The agricultural revolution, which began around 10,000 BCE, marked the transition from mobile communities of hunters to settled agricultural communities. Innovations (such as the domestication of plants and animals, the invention of the plough, and the development of irrigation systems) made it possible to produce surplus food, which supported larger populations and allowed the establishment of permanent settlements.

The Industrial Revolution, which started in the eighteenth century, marked a transition from agrarian to industrial economies and the emergence of urban communities. Powered by advancements in steam power, mechanisation, and the techniques of mass production, industries (such as textiles, manufacturing, and transportation) underwent radical transformations. The invention of the steam engine, the spinning jenny, and the power loom revolutionised

production processes, leading to unprecedented levels of productivity and economic growth. The Industrial Revolution also fuelled urban migration, social upheaval, and the rise of capitalism, shaping the modern world in profound ways. The mechanisation of agriculture in the late nineteenth century and early twentieth century represents another significant technological disruption because it displaced millions of farm workers, contributing to rural depopulation and migration to urban centres in search of employment opportunities.

The digital revolution, which began in the mid-twentieth century, marked the advent of computing, telecommunications, and digital technologies. A new era of technological disruption began with the introduction of innovations such as the personal computer, the internet, and mobile technologies, leading to a radical transformation of communication, commerce, and entertainment. While these developments allowed more efficient production techniques, they also raised concerns about job displacement. Breakthroughs (such as the invention of the transistor, the development of integrated circuits, and the advent of the internet) transformed the way information is processed, stored, and transmitted. This wave of technological progress also allowed the rise of new industries (such as software development, telecommunications, and e-commerce) that propelled economic growth and globalisation.

The emergence of AI represents the latest chapter in the story of technological disruption. Technological breakthroughs have enabled AI systems to perform tasks that were thought to be exclusive to human intelligence. AI is transformative and disruptive because it is reshaping industries, professions, and modes of work. The historical precedents of technological disruption offer valuable insights into the potential implications and challenges posed by AI. For example, it has led to the emergence of a new class of the super-rich, the high-tech billionaires. While disruptive technologies have historically spurred economic growth and prosperity, they have also exacerbated inequality, displaced traditional forms of labour, and reshaped social structures. All of these characteristics are valid for AI. However, Eric Schmidt (a former CEO of Google) believes that “AI is likely to be the most transformative technology of our time” (Choat, 2023). In other words, this time it is different!

AI represents a disruptive innovation that challenges existing paradigms, processes, and industries because it has reshaped the way people work, communicate, and live. AI drives economic transformation by enhancing productivity, creating new industries, and fostering innovation. AI-powered applications in areas such as healthcare, finance, and transportation create new opportunities for growth, efficiency, and competitiveness, driving economic expansion and prosperity. AI has profound social implications, including job displacement, skills mismatch, and income inequality. Gruetzemacher and Whittlestone (2022) examine the transformative power of AI, arguing that “transformative AI” reflects “the possibility that advanced AI systems could have very large

impacts on society without reaching human-level cognitive abilities.” They go on to distinguish among narrowly transformative AI, transformative AI, and radically transformative AI.

While comparisons between AI and previous technological revolutions reveal similarities in terms of disruptive innovation, economic transformation, and social impacts, AI exhibits unique characteristics, including its global scale, rapid pace of change, and human-centric nature. An ongoing debate centres on the power of AI to displace labour and create mass unemployment. Most observers adopt the attitude of “no worries” on the grounds that new technologies create more jobs than the jobs they destroy, and AI is no exception. However, there are those who believe that this time it is different and that we should start seriously thinking about how to respond to the disruptive power of AI. For example, Elon Musk believes that no job is immune from the effect of AI, which will make it necessary for governments to pay their citizens universal basic income. We will come back to this issue and discuss it in detail in Chapter 6.

1.8 EVOLUTION OF AI: SUMMING UP

The evolution of AI is characterised by waves of technological advancement, each wave marking a significant milestone in the evolution of intelligent systems. From a humble beginning in the 1950s, the field of AI has gone through significant transformations, propelled by breakthroughs in computing power, algorithmic innovations, and the availability of big data. We have gone through a detailed chronology of the evolution of AI, starting from ancient times. In this concluding section, we highlight key developments, paradigm shifts, and the impact on society, the economy, and human-machine interaction. With respect to paradigm shifts, Dhar (2023) summarises the evolution of AI in terms of shifts from expert systems to machine learning to deep learning and finally to general-purpose technology.

The two decades of the 1950s and 1960s witnessed the first wave of AI, which was rooted in symbolic logic and rule-based systems. The objective of symbolic logic was to encode human knowledge and expertise to formulate rules and representations that enabled computers to perform tasks involving logical reasoning and problem-solving. A prominent application of symbolic AI was the construction of expert systems that emulated the decision-making ability of human experts in specific fields such as medicine, finance, and engineering. However, expert systems lacked the ability to learn from data or adapt to new situations.

The second wave of AI, observed in the 1980s and 1990s, is known as the “connectionist revolution,” where connectionism is an approach to AI that developed out of attempts to understand how the human brain works at the

neural level—in particular, how people learn and remember. Connectionist models of language resemble, in one way or another, the central nervous system of humans. Inspired by the structure and functioning of the human brain, researchers explored the use of artificial neural networks for pattern recognition, classification, and function approximation. Breakthroughs such as the backpropagation algorithm and the development of deep learning reignited interest in neural networks, paving the way for their application in speech recognition, image processing, and natural language understanding. However, limitations in computational resources and algorithmic scalability hindered widespread adoption during this period.

The third wave of AI began in the early twenty-first century, propelled by a combination of big data, powerful computers, and advances in deep learning that led to breakthroughs in image recognition, natural language processing, and game playing. This wave of AI witnessed the rise of AI-driven applications in diverse domains, including autonomous vehicles, healthcare diagnostics, and personalised recommendation systems.

The fourth wave of AI is characterised by the integration of AI with human-centric technologies. For example, Internet of Things (IoT) devices enable AI to operate in real time, empowering applications such as smart cities, industrial automation, and wearable devices. Furthermore, emphasis has shifted to human-centric AI, where intelligent systems are designed to collaborate with humans, augmenting human capabilities rather than replacing them. This wave of AI also raises ethical, societal, and regulatory considerations, prompting discussions on transparency, accountability, and bias mitigation. We will come back to this issue in Chapter 5 when we discuss the adverse consequences of AI.

As AI continues to evolve and permeate various aspects of our lives, it is essential to approach its development and deployment with caution, ensuring that it serves the collective good while mitigating potential risks. By understanding the historical context, conceptual frameworks, and implications of AI, we can navigate its complexities and harness its transformative potential for the betterment of humanity.